

Synology DS725+ NAS — complete setup guide

Personal reference covering hardware choices, Tailscale VPN setup, Synology Photos remote access, and everything learned along the way.

1. Hardware

NAS: Synology DS725+ (~\$430 / €399)

- AMD Ryzen R1600 dual-core, 2.6 GHz (turbo 3.1 GHz)
- 4GB DDR4 ECC RAM (expandable to 32GB)
- 2 × 3.5"/2.5" SATA bays + 2 × M.2 NVMe slots (PCIe 3.0)
- 1 × 2.5GbE + 1 × 1GbE Ethernet
- Expandable to 7 drives via DX525 expansion unit
- Runs DSM 7.3 — supports Synology Photos, Tailscale, and all required packages
- Noise: 17.9 dB(A) — quieter than a whisper

HDDs: Seagate IronWolf 4TB (ST4000VN006) × 2

- CMR recording (RAID-safe, no SMR write issues)
- 5400 RPM, 256MB cache
- 180TB/year workload rating
- 1,000,000 hours MTBF
- 3-year warranty + 3 years Seagate Rescue Data Recovery
- RV sensors for vibration tolerance (4TB and above)
- IronWolf Health Management integrates with Synology DSM

DS725+ compatibility: The 2025 Synology "Plus" models initially restricted third-party drives, but the DSM 7.3 update (October 2025) reversed this via the HCL 5.0 framework. Seagate IronWolf drives now work without warnings or restrictions on the DS725+ — pool creation, RAID builds, and expansion all function normally. Coolblue and other retailers even sell the DS725+ bundled with IronWolf 4TB drives. Make sure your DS725+ is running DSM 7.3 or later before installing third-party drives.

RAID configuration: RAID 1 (mirroring)

- 2 × 4TB drives → ~4TB usable capacity
- Both drives hold identical copies of all data
- One drive can fail without data loss
- Read performance can be slightly faster (reads from both drives)

Optional: SSD read cache

The DS725+ has $2 \times$ M.2 2280 NVMe slots (PCIe 3.0). Adding an SSD cache speeds up access to frequently used files — helpful for browsing Synology Photos thumbnails, running searches, and general DSM responsiveness. It does NOT speed up large sequential transfers like video streaming.

SSD recommendation — Synology SNV3410 400GB (~\$150)

The DS725+'s M.2 slots may restrict caching to Synology-branded NVMe drives (SNV3400/ SNV3410 series). Before buying a third-party NVMe like the Samsung 970 EVO Plus, check the compatibility list for the DS725+ specifically. The SNV3410 offers 3,000 MB/s reads, 750 MB/s writes, and 491 TBW endurance — more than enough for a read cache.

For a **read-only cache** (recommended), one SSD is sufficient. No risk of data loss if the cache SSD fails — DSM falls back to reading from the HDDs. Uses ~416KB of system RAM per 1GB of SSD cache.

2. RAID 1 drive upgrade process (e.g., $2 \times 4\text{TB} \rightarrow 2 \times 8\text{TB}$)

This can be done with zero downtime and no data loss.

Before starting:

- Back up your data (even though RAID 1 protects against drive failure)
- Confirm the storage pool status is "Normal" in Storage Manager
- Run a Btrfs file system scrub if applicable
- Check that new drives are on the DS725+ compatibility list

Drive 1:

1. Open Storage Manager → Storage Pool → select the first drive → **Deactivate**
2. The drive bay LED turns amber — confirms which physical bay to pull
3. Remove the 4TB drive, insert the new 8TB drive
4. In Storage Manager, **Repair** the storage pool
5. Wait for rebuild to complete (several hours) — data stays accessible during this time
6. Wait until status returns to **Healthy**

Drive 2: 7. Repeat: deactivate → swap → repair for the second drive 8. This time, the repair dialog shows an option to **Expand capacity** — check that box 9. Wait for repair + expansion to complete

After: Verify both 8TB drives show healthy and full capacity is available. Critical rule: **never replace both drives at the same time** — always rebuild fully between replacements.

3. Tailscale setup for secure remote access

Network architecture

Show Image

Key points:

- Your data flows **directly** between phone and NAS (peer-to-peer)
- Tailscale coordination servers only handle device discovery — they cannot read your data
- Your home router keeps **all ports closed**
- All traffic is end-to-end encrypted with WireGuard (keys only on your devices)

Step-by-step setup

A. Install Tailscale on the NAS

1. Log in to DSM → **Package Center** → search for **Tailscale** → **Install**
 - If not in Package Center: download the correct `.spk` from <https://pkgs.tailscale.com/stable/#spks> (DS725+ uses **x86_64**)
 - Package Center → **Manual Install** → upload the `.spk`
2. Open Tailscale → **Log In** → authenticate with Google/Microsoft/GitHub
3. Note your NAS's Tailscale IP (e.g., `100.64.0.1`) from the admin console at <https://login.tailscale.com/admin/machines>

B. Enable outbound connections (required for DSM 7)

1. **Control Panel** → **Task Scheduler** → **Create** → **Triggered Task** → **User-defined script**
2. General tab: name it `Tailscale TUN Setup`, user: `root`, event: `Boot-up`, enabled: ✓
3. Task Settings tab, paste:

```
/var/packages/Tailscale/target/bin/tailscale configure-host; synosystemctl  
restart pkgctl-Tailscale.service
```

4. Save → **reboot the NAS** (or right-click → Run)

C. Advertise your local subnet (optional, recommended)

This lets you access all home network devices through Tailscale using their normal local IPs.

1. **Control Panel** → **Task Scheduler** → **Create** → **Scheduled Task** → **User-defined script**
2. User: `root`, schedule: do not repeat (run manually once)
3. Task Settings, paste (adjust subnet to match yours):

```
/var/packages/Tailscale/target/bin/tailscale up --advertise-routes=192.168.1.0/24 --advertise-exit-node --reset
```

4. Right-click → **Run**
5. Go to <https://login.tailscale.com/admin/machines> → find your NAS → three dots → **Edit route settings** → enable the subnet route

D. Install Tailscale on your iPhone

1. Download **Tailscale** from the App Store
2. Sign in with the **same identity provider** used for the NAS
3. Allow the VPN configuration when prompted
4. Toggle Tailscale **on** — your phone is now on the same tailnet as your NAS

E. Connect Synology Photos via Tailscale

1. Ensure Tailscale is connected on your iPhone
2. Open the **Synology Photos** app
3. Log in using your NAS's Tailscale IP: `https://100.x.x.x:5001`
 - Or use the MagicDNS name: `https://your-nas.your-tailnet.ts.net:5001`
4. Enter your DSM credentials
5. Enable **Photo Backup** in the app's settings

F. Disable QuickConnect (optional, recommended)

Control Panel → **QuickConnect** → **uncheck Enable QuickConnect** → **Apply**

Your NAS is now only reachable via your local network or Tailscale.

G. Firewall exception (if DSM firewall is enabled)

Control Panel → **Security** → **Firewall** → **Edit Rules** → add:

- Source IP: `100.64.0.0` / Mask: `255.192.0.0`
- Ports: All
- Action: Allow
- Place this rule **above** any deny-all rule

4. VPN On Demand (iOS) — auto-connect only when needed

Instead of keeping Tailscale always-on, you can configure iOS to automatically activate the VPN only when you try to reach a `.ts.net` address.

Setup

1. Open Tailscale on iPhone → tap profile picture → **VPN On Demand**
2. Set **Wi-Fi** to **Do Nothing**
3. Set **Cellular** to **Do Nothing**
4. Enable **MagicDNS matching** (`*.ts.net`)

How it works

- VPN stays off during normal browsing
- When any app tries to reach `*.ts.net`, iOS automatically activates Tailscale
- Tunnel establishes in ~2 seconds, then the connection goes through
- When traffic stops, the VPN can be manually disconnected

Important: use MagicDNS names

For On Demand to trigger, Synology Photos must connect using the MagicDNS hostname (not a raw IP):

```
https://your-nas.your-tailnet.ts.net:5001
```

Find your NAS's MagicDNS name in the Tailscale admin console under Machines.

Trade-off

On Demand is great for occasional access. But for **continuous photo backup**, always-on is better — it lets Synology Photos sync new photos in the background without waiting for you to trigger the VPN.

5. How VPN tunnels actually work (quick reference)

The connection lifecycle

1. **Registration** — Both devices make outbound connections to Tailscale's coordination server. They share their public keys and NAT-assigned public IP:port. Outbound connections always work through NAT without open ports.
2. **NAT hole punching** — The coordination server tells each device the other's public IP:port. Both simultaneously send UDP packets to each other, causing both routers to create allow rules for the "return" traffic. Direct path is now open.
3. **Direct tunnel** — WireGuard encrypts everything end-to-end. Actual data path: Phone → LTE → internet → ISP → router → NAS. Tailscale servers are completely out of this path.
4. **Keepalives** — WireGuard sends a tiny (100-byte) packet every 25 seconds to prevent NAT mappings from expiring. Cost: ~350KB/day.
5. **Fallback** — If hole punching fails (rare), Tailscale routes through DERP relay servers. Data is still WireGuard-encrypted end-to-end — the relay sees only opaque bytes and cannot decrypt anything.

Network changes

Event	Recovery
Phone switches Wi-Fi ↔ LTE	Re-punch in 1-3 seconds, automatic
Router reboots	NAT table clears, re-punch automatic
NAS reboots	Boot-up script runs, re-registers, resumes
Internet drops briefly	WireGuard is stateless — resumes instantly

6. Security layers

Layer 1: Tailscale identity (SSO via Google/Microsoft + device approval)
Layer 2: WireGuard encryption (Curve25519 + ChaCha20 + Poly1305)
Layer 3: DSM authentication (username + password + optional 2FA)
Layer 4: Synology Photos permissions (per-user access controls)
Layer 5: No open ports on router (zero public attack surface)

What's protected

- **Data in transit** — WireGuard E2E encryption; ISP, carrier, coffee shop Wi-Fi can't read anything
- **Against coordination server compromise** — Attacker can't decrypt traffic (no WireGuard private keys)
- **Against MITM** — WireGuard authenticates both endpoints with public/private key pairs

What to watch for

- **Endpoint security** — If your phone or NAS is compromised, the VPN doesn't help. Use strong passwords, 2FA on DSM, keep DSM updated.
- **Identity provider** — If your Google/Microsoft account is compromised, attacker could add a device to your tailnet. Use 2FA on your identity provider.
- **Tailscale trust** — You trust Tailscale for key exchange brokering. For zero third-party trust, self-host with Headscale (open-source coordination server).
- **NVMe M.2 restriction** — DS725+ may require Synology-branded SSDs (SNV3400/ SNV3410) for caching. Check before buying third-party NVMe.
- **HDD prices** — Storage prices have risen ~46% since September 2025 due to AI infrastructure demand. Shop around.

7. Shopping list

Item	Model	Est. price	Notes
NAS	Synology DS725+	~\$430	2-bay, expandable to 7
HDD × 2	Seagate IronWolf 4TB (ST4000VN006)	~\$90 each	CMR, 3yr warranty + Rescue
SSD cache (optional)	Synology SNV3410 400GB	~\$150	PCIe 3.0, read-only cache
UPS (recommended)	Any USB-compatible UPS	varies	Protects against power loss during writes

Total (without cache): ~\$610 Total (with cache): ~\$760